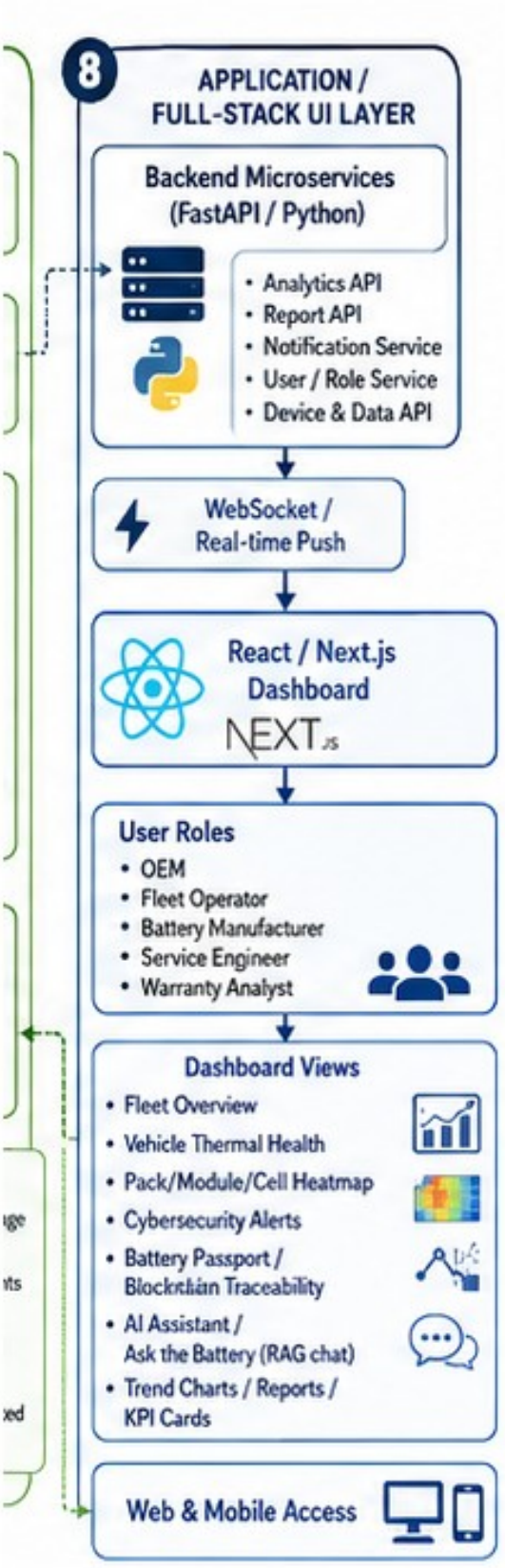
Step 8 - Website / Full-Stack UI Layer

Website-Only Development, Testing and Acceptance Guide



Development Workflow | Testing & Validation | Acceptance Criteria | Timeline | Reference Material

Revision 1 | August 2026

Document Purpose and Structure

This guide converts the platform architecture block into an implementable engineering work package. It defines the functional scope, interfaces, development activities, verification strategy, acceptance criteria, evidence required for sign-off, delivery timeline, and reference resources.

All numerical acceptance values in this guide are baseline project targets for the first controlled release. They should be recalibrated after representative load testing, vehicle/telematics characterization, and customer/OEM requirements are frozen.

Document structure

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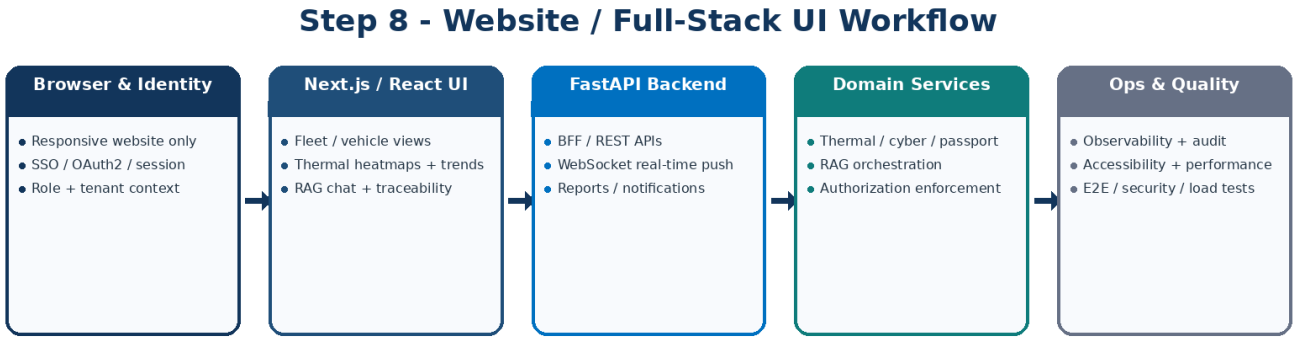
1. Website Scope and User Roles

Step 8 is the customer-facing website and API orchestration layer for the Battery Thermal Intelligence Cloud Platform. The current release is explicitly website-only. No native Android/iOS application, app-store packaging or mobile-specific backend is included.

Current scope: responsive web application accessible from supported desktop/tablet/mobile browsers. Native mobile APP development is out of scope for this release.

User role	Primary needs	Website access
OEM engineering	Fleet health, detailed thermal/cyber/passport diagnostics, trends and evidence	Full tenant engineering views
Fleet operator	Vehicle risk ranking, alerts, uptime/maintenance priorities	Fleet operational views
Battery manufacturer	Pack/module performance, field issue patterns, passport traceability	Battery/product views
Service engineer	Vehicle diagnosis, hotspot/cooling/sensor faults, recommended checks	Case/service views
Warranty analyst	Historical events, thermal exposure, similar cases, evidence	Warranty analytics views
Tenant administrator	Users/roles, devices, access policy, subscriptions/configuration	Administration views

2. End-to-End Full-Stack Workflow



1. User opens the responsive website; identity provider authenticates the user and returns tenant/role context.
2. Next.js/React loads role-appropriate navigation and requests summary APIs through a Backend-for-Frontend (BFF) or FastAPI service.
3. Backend enforces authorization again and calls thermal, cyber, blockchain/passport and RAG services using service identities.
4. Real-time telemetry, alerts and status changes are delivered through a controlled WebSocket/SSE channel; historical data uses REST APIs.
5. Website renders fleet cards, vehicle health, thermal maps, trends, cyber alerts, passport traceability and RAG answers with source citations.
6. Report/export actions are generated server-side with audit records; sensitive data never relies on browser-side authorization alone.
7. Metrics, traces, frontend errors and audit events are collected for operational monitoring and product analytics.

3. Information Architecture and Pages

Page / module	Core content
Login / tenant selector	SSO/OIDC, allowed tenants/roles, session and MFA flow
Fleet Overview	Fleet health score, thermal/cyber risk counts, priority vehicles, filters
Vehicle Detail	Battery identity, current state, thermal KPIs, recent alerts, trend charts
Pack / Module / Cell Heatmap	Interactive temperature map, Tmax/Tmin/Tavg/DeltaT, hotspot selection
Thermal Intelligence	Hotspot, cooling/sensor health, forecast/time-to-limit, recommendations
Cybersecurity Alerts	Trust status, quarantined streams, alert/evidence timeline
Battery Passport	Lifecycle events, provenance, verification status, blockchain anchor
Ask the Battery (RAG)	Natural-language query, cited sources, related cases, evidence cards
Reports	Downloadable fleet/vehicle/warranty/compliance reports

Page / module	Core content
Administration	Users, roles, devices, tenant settings, alert preferences

4. Website Architecture

Layer	Baseline technology / responsibility
Browser	Modern desktop/tablet/mobile browser; no native app
CDN / reverse proxy / WAF	TLS termination, static asset caching, rate/security controls
Frontend	Next.js + React + TypeScript; responsive UI; server/client rendering as appropriate
BFF / Backend APIs	FastAPI/Python; session/tenant context; aggregation of domain APIs
Real-time channel	WebSocket or Server-Sent Events for alert/telemetry updates
Domain services	Thermal intelligence, cyber trust, blockchain/passport, RAG, reports
Persistence/cache	PostgreSQL/time-series stores, Redis/cache where justified
Observability	OpenTelemetry traces/metrics/logs, frontend error monitoring, audit events

5. Backend Microservices / BFF

- Analytics API: fleet, vehicle, pack, module and cell KPI aggregation.
- Report API: asynchronous report request/status/download with access checks.
- Notification service: alert inbox and optional email/web notifications; no native push app dependency.
- User/role service: tenant-scoped identity/profile mapping; preferably integrate external IdP rather than store passwords locally.
- Device/data API: normalized metadata and historical telemetry queries.
- RAG proxy/orchestrator API: authenticated question, contextual retrieval, streaming answer and citation objects.
- Passport API: event timeline, verification status and anchor proof summaries.

6. REST APIs and WebSocket Real-Time Channel

Interface	Purpose	Key controls
REST GET/POST	Historical queries, reports, admin/config	JWT/session auth, tenant/RBAC, pagination, validation, rate limits
WebSocket/SSE	Live alerts, KPI changes, thermal/cyber events	Authenticated channel, tenant topics, reconnect/heartbeat, bounded message size
Report download	Generated PDF/CSV artifacts	Short-lived signed URL or authenticated stream; audit download
RAG streaming	Progressive answer tokens + citations	Session isolation, cancellation, timeout, content policy
Admin APIs	Users/roles/devices/preferences	Elevated RBAC, audit, CSRF/session protections

Authorization must be enforced on every backend request and subscription. Hiding a UI menu is not an access-control mechanism.

7. React / Next.js Frontend

- Use a component design system for KPI cards, tables, alerts, charts, heatmaps and source/evidence cards.
- Prefer TypeScript types generated from OpenAPI schemas to reduce frontend/backend contract drift.
- Use route-level and component-level loading/error states; never leave stale thermal/cyber status without freshness indication.
- For dense engineering charts, provide zoom, tooltip, selectable time range, unit labels and downloadable data where permitted.
- Persist user filters only where safe; do not store access tokens or sensitive telemetry in localStorage unless security design explicitly permits it.
- Use server rendering/static rendering where it improves first load; real-time pages can hydrate and subscribe after authentication.

8. Authentication, Session and RBAC

Control	Implementation expectation
Login	OIDC/OAuth2 via enterprise IdP; MFA according to customer policy
Session	Secure, HttpOnly, SameSite cookies or equivalent hardened token handling
RBAC	Tenant + role + resource-level checks in backend
Tenant isolation	Every query and WebSocket subscription scoped by authenticated tenant
CSRF/XSS	Framework protections plus CSP/output encoding and security testing

Control	Implementation expectation
Audit	Login/logout, admin changes, report exports, sensitive detail access recorded
Timeout	Idle and absolute session expiry; explicit reauthentication for sensitive admin actions

9. Thermal and Fleet Visualization

Visualization	Required behavior
Fleet KPI cards	Healthy/Monitor/High/Critical counts with drill-down and filter context
Vehicle thermal health	Tmax, Tavg, DeltaT, dT/dt, risk score, cooling/sensor health, freshness
Cell/module heatmap	Color scale with numeric tooltip, hotspot marker, physical-vs-virtual indication
Trend chart	Measured/virtual temperature, current, SOC, cooling command, alerts on aligned time axis
Forecast	Current Tmax plus T+5/T+10/time-to-limit and confidence/assumption indicator
Recommendations	Action, severity, evidence/reason and owner/status; clearly separate recommendation from completed action

10. RAG Assistant Website Experience

- Chat panel must show answer text, source cards/citations, timestamp/model context and an explicit insufficient-evidence state.
- Users should be able to ask questions from a fleet/vehicle context without manually repeating IDs; context must remain tenant-authorized.
- Citation click opens approved document metadata and, where supported, exact page/section.
- Provide suggested questions such as Why is Module 4 overheating?, Show similar warranty cases, and Which procedure applies?.
- Display that RAG output is advisory and evidence-grounded; do not present generated text as an executed control action.
- Capture thumbs-up/down and issue feedback linked to query/retrieval trace for evaluation.

11. Battery Passport / Blockchain Traceability Views

- Timeline of lifecycle/service/compliance/critical events with event type, time, actor and source.
- Show off-chain record status plus blockchain anchor/transaction proof summary without overwhelming normal users with low-level ledger details.
- Expose verify action to recalculate/confirm record hash where the backend supports it.
- Use role-based masking for confidential supplier/service data.
- Provide QR/battery-passport identifiers and exportable evidence only according to approved policy.

12. Reports, Notifications and Exports

Function	Baseline behavior
Alert inbox	Filter by severity, vehicle, subsystem, status and time; acknowledge/assign where workflow permits
Email/web notification	Optional notification for configured severity; links back to authenticated website
PDF report	Server-generated summary with timestamp, tenant, filters and data freshness
CSV export	Permission-controlled raw/processed data export with row/time limits
Warranty case pack	Selected events, trends, RAG citations, service history and provenance
Audit	Every sensitive export/report records requesting user, scope and time

13. Responsive Web, Accessibility and UX

Website-only does not mean desktop-only. The release should be responsive for supported tablet/mobile browsers, but no native app binaries, app stores, native push notifications or device-specific SDKs are included.

- Target WCAG 2.2 AA for keyboard navigation, focus visibility, contrast, semantic structure and accessible names.
- Do not communicate severity only by color; include text/icon labels.
- Heatmaps need numeric values/tooltips and legends that remain usable for color-vision deficiencies.
- Use consistent units, time zones and freshness indicators throughout the site.
- Define supported browser matrix and minimum viewport sizes before UAT.
- Provide empty/error/loading/offline states and clear retry behavior.

14. Deployment, Observability and Operations

- Containerize frontend/backend and use environment-specific configuration with secret management.
- CDN/cache immutable static assets; do not cache tenant-sensitive API responses publicly.
- Collect backend traces/metrics/logs and frontend JavaScript errors/session performance without collecting unnecessary sensitive content.
- Monitor API latency/error rate, WebSocket connections, page Core Web Vitals, report/RAG failures and authorization denials.
- Use feature flags for staged release of new dashboards or RAG features.
- Support blue/green or rolling deployment plus rapid rollback for frontend/backend independently.

15. Development Lifecycle and Completion Timeline

Activity	Deliverable	Target timing	Exit condition
User roles, journeys and page map	Information architecture + wireframes	Week 1	Stakeholders approve website scope
Design system / UX baseline	Components, layout, responsive breakpoints	W1-W2	Core UI patterns approved
API contract freeze	OpenAPI/WebSocket contracts	W2-W3	Backend/frontend mocks pass
Authentication/RBAC skeleton	Login/session/tenant route guards	W2-W4	Positive/negative access flows pass
Backend/BFF services	Analytics/device/report APIs	W3-W6	API unit/contract tests green
Fleet/vehicle dashboards	KPI, trends, filters, alert views	W4-W7	Representative data rendered correctly
Thermal heatmap/intelligence views	Pack/module/cell visualization	W5-W8	Engineering review accepts semantics
Cyber/passport views	Alert + provenance/ledger pages	W6-W9	Step 3/6 API integration pass
RAG website experience	Chat streaming + citations	W7-W10	Step 7 integration + security pass
Reports/exports	PDF/CSV/report workflow	W8-W10	Permission and audit tests pass
Accessibility/performance hardening	WCAG, Core Web Vitals, browser fixes	W10-W12	Quality targets met
Security/load/E2E regression	Full release test suite	W11-W13	No release-blocking findings
Customer/UAT pilot	Representative OEM/fleet/service users	W13	UAT closure complete
Production release	Runbook, monitoring, rollback, SBOM, sign-offs	End W14	G7 approval

16. Verification and Validation Strategy

V&V layer	Method	Evidence
Frontend unit/component	Component states, formatting, heatmap/legend, role visibility	Automated unit/component report
Backend unit/API	Validation, authz, pagination, aggregation, error handling	Coverage + API report
Contract	OpenAPI/schema and WebSocket message compatibility	Contract test report
E2E	Login -> fleet -> vehicle -> heatmap -> RAG/report flows	Playwright/Cypress regression
Security	OWASP ASVS/API Top 10, RBAC/tenant,	Security report

V&V layer	Method	Evidence
	XSS/CSRF/session	
Accessibility	Automated + keyboard/screen-reader spot checks	WCAG audit
Performance	Core Web Vitals, API/WebSocket load, large heatmap/trends	Performance report
Compatibility	Supported browsers and responsive viewport matrix	Browser matrix report
UAT	OEM/fleet/service/warranty representative tasks	UAT sign-off

17. Test Catalogue

Test ID	Scenario	Expected result
WEB-01	Valid OEM login	Correct tenant/role dashboard displayed
WEB-02	User attempts another tenant URL/API	Access denied; no data disclosed
WEB-03	Fleet filter/drill-down	KPIs and list use identical filter context
WEB-04	Vehicle thermal page	Values/units/freshness match API source
WEB-05	Heatmap hotspot	Correct cell/module highlighted with numeric tooltip
WEB-06	WebSocket reconnect	Session resumes without duplicate alert cards
WEB-07	RAG answer with citations	Sources visible and resolvable
WEB-08	RAG no-evidence query	Insufficient-evidence state rendered clearly
WEB-09	Passport event timeline	Events ordered and verification status correct
WEB-10	Report export	Authorized report generated; export audited
WEB-11	Expired session	Protected route/API requires reauthentication
WEB-12	XSS/HTML payload in source field	Rendered safely; no script execution
WEB-13	Keyboard-only navigation	All critical flows operable with visible focus
WEB-14	Mobile browser width	Responsive layout usable; no horizontal clipping of core views
WEB-15	Peak concurrent users	Latency/error/WS stability within target
WEB-16	Backend/RAG unavailable	Graceful degraded/error state; no stale success indication

18. Acceptance Criteria - Project Targets

Metric	Baseline project target	Acceptance method
RBAC / tenant isolation	100% of negative access tests denied	Automated API/E2E security tests
Critical/High security findings	0 open at release	SAST/SCA/DAST/manual closure
API availability in pilot	>= 99.9% during agreed pilot service window	Monitoring report
API latency	P95 <= 750 ms for standard non-RAG read APIs at nominal load	Load test
WebSocket event delivery	P95 server event -> browser receipt <= 2 s at nominal load	Instrumented real-time test
RAG interactive response	First visible streamed response <= 3 s P95; total per Step-7 acceptance	E2E load test

Metric	Baseline project target	Acceptance method
Largest Contentful Paint	<= 2.5 s at P75 on agreed test profile for primary pages	Lighthouse/Web Vitals test
Interaction to Next Paint	<= 200 ms at P75 on agreed test profile	Web Vitals test
Cumulative Layout Shift	<= 0.1 at P75	Web Vitals test
Accessibility	No unresolved WCAG 2.2 AA blockers in agreed audit scope	Automated + manual audit
Browser support	100% critical UAT flows pass on approved browser matrix	Cross-browser E2E
Data correctness	100% sampled KPI/heatmap/report values trace to API/source calculation	Traceability audit
Report/export authorization	100% unauthorized export tests denied and authorized exports audited	Security/audit tests

19. Acceptance Gates and Sign-off

Gate	Review question	Required evidence
G0 - UX/Scope Freeze	Are website-only scope, roles, pages and out-of-scope native app explicit?	IA, wireframes, scope statement
G1 - Contract Freeze	Are APIs, WebSocket messages, auth/session and error contracts stable?	OpenAPI/WS contracts
G2 - Design System Ready	Are components, responsive behavior, accessibility rules and data semantics defined?	Design system + prototypes
G3 - Functional Integration	Do thermal, cyber, passport, RAG and report views integrate correctly?	Integration/E2E evidence
G4 - Security/Privacy	Do RBAC, tenant, session, XSS/CSRF/API-security tests pass?	Security report
G5 - Quality Acceptance	Do accessibility, performance, browser and data-correctness targets pass?	Quality scorecards
G6 - Customer UAT	Do target users complete critical workflows successfully?	UAT report and closure
G7 - Production Release	Are deployment, monitoring, rollback, SBOM, runbooks and sign-offs complete?	Release pack

20. Risks and Mitigations

Risk	Impact	Mitigation
UI/API contract drift	Broken pages or wrong values	OpenAPI-generated types + contract CI
Dense data overwhelms browser	Slow charts/heatmaps	Aggregation/downsampling, virtualization, bounded time ranges
Cross-tenant frontend bug	Potential confidentiality breach	Backend enforcement + isolation tests; never trust UI filtering
Real-time reconnect duplicates	Repeated alerts / inconsistent status	Event IDs, resume cursor, idempotent UI store
RAG latency	Poor experience	Streaming, loading states, caching/routing per Step 7
Accessibility added late	Expensive rework	Design-system accessibility from Week 1
Native-app expectations creep	Scope/timeline expansion	Explicit website-only release boundary and backlog item for future app
Browser/device variability	Layout/feature defects	Approved browser/viewport matrix and E2E tests

21. Engineering Deliverables

- Website information architecture and user-role matrix
- Wireframes/design system and responsive breakpoints
- OpenAPI and WebSocket interface specifications
- Next.js/React frontend source and component library
- FastAPI/BFF backend services and API tests
- Thermal/cyber/passport/RAG integration adapters
- Heatmap/trend/KPI visualization specifications
- Authentication/session/RBAC implementation and matrix
- Report/export and audit workflow
- Automated frontend/backend/contract/E2E/security/accessibility/load tests
- Browser compatibility and Web Vitals report
- Customer UAT report
- Deployment manifests, SBOM, monitoring dashboards, runbook and rollback plan
- Production gate sign-offs

22. Reference Material and Open-Source Validation Resources

[1] Next.js Official website framework documentation

[2] React Official React documentation

[3] FastAPI Official Python web API framework documentation

[4] Playwright Open-source end-to-end browser testing

[5] Grafana k6 Open-source load/performance testing

[6] OpenTelemetry Observability standard and SDKs

[7] W3C Web Content Accessibility Guidelines (WCAG) 2.2

[8] web.dev Core Web Vitals guidance

[9] OWASP ASVS Application Security Verification Standard

[10] OWASP API Security API Security Top 10

[11] Zync: A microservice platform for IoT device management and complex event monitoring Computers & Electrical Engineering, 2025

[12] Harnessing digital twin and IoT for real-time monitoring, diagnostics, and error correction in domestic solar energy storage Energy Reports, 2024

[13] An Enhanced Cloud-Based Digital Twin State Monitoring for Smart Stationary Battery Energy Storage System IEEE SMART 2024

[14] Explainable Data-Driven Digital Twins for Predicting Battery States in Electric Vehicles IEEE Access, 2024

[15] Development of end-to-end low-cost IoT system for densely deployed PM monitoring network: an Indian case study Frontiers in the Internet of Things, 2024

[16] SmartMonitor Open-source MQTT + FastAPI + real-time web dashboard reference project